

DATA ANALYTICS INTERNSHIP

-InternSpark

(AICTE-Listed Internship Program)

REPORT ON

Website Traffic Analysis Report

(TASK 3)

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Introduction

Website traffic analysis is essential for understanding user behavior, engagement patterns, and content performance. The objective of this project is to analyze the website traffic dataset and extract meaningful insights that can support data-driven business decisions.

The analysis focuses on event activity, geographical distribution of users, popular artists, albums, tracks, and traffic trends over time. Various data visualization techniques have been used to present the findings in a clear and understandable manner.

Objective

The primary objectives of this project are:

- To clean and preprocess the website traffic dataset.
- To identify overall traffic patterns.
- To analyze user activity across countries and cities.
- To determine the most popular artists, albums, and tracks.
- To study event distribution.
- To analyze traffic trends over time.
- To provide actionable business recommendations.

Dataset Overview

The dataset contains website traffic and user interaction information including:

- Event
- Date
- Country
- City
- Artist
- Album
- Track
- ISRC

This data helps in understanding user engagement and content consumption behavior.

Data Cleaning and Preprocessing

Before performing the analysis, the dataset was cleaned to ensure accuracy and reliability.

Missing Values

Missing values were identified and examined.

Duplicate Removal

Duplicate records were detected and removed to avoid inaccurate analysis.

Date Conversion

The Date column was converted into datetime format for time-series analysis.

After preprocessing, the dataset became suitable for analysis and visualization.

The following metrics were calculated:

Key Performance Metrics

- Total Events
- Number of Unique Countries
- Number of Unique Cities

These metrics provide an overview of the traffic dataset.

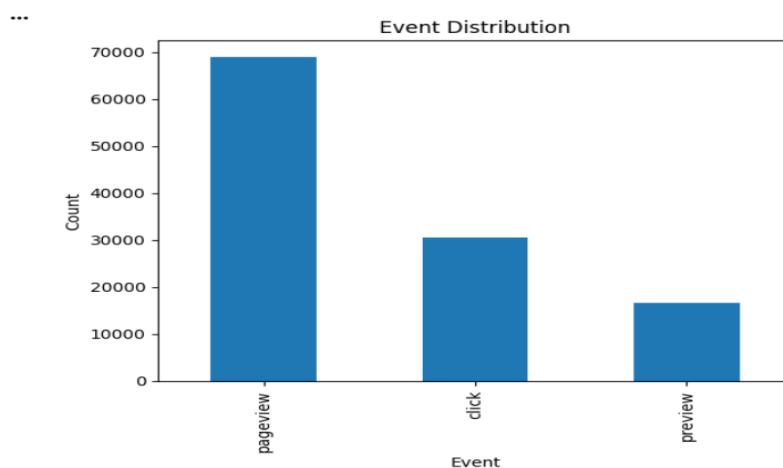
Data Visualization and Analysis:-

Event Distribution

A bar chart was created to analyze the distribution of events.

Insights:

- Certain event types occurred more frequently than others.
- User activity is concentrated around specific interactions.



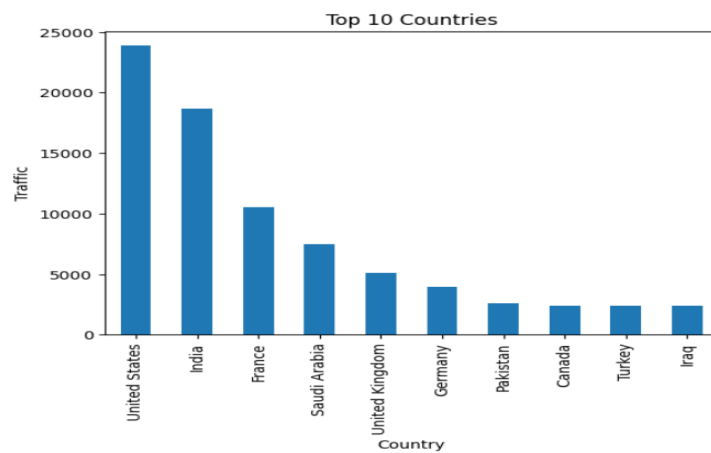
Top 10 Countries

A bar chart was used to identify countries generating the highest traffic.

Insights:

- Traffic is concentrated in a few major countries.

- High-performing countries represent key target markets.

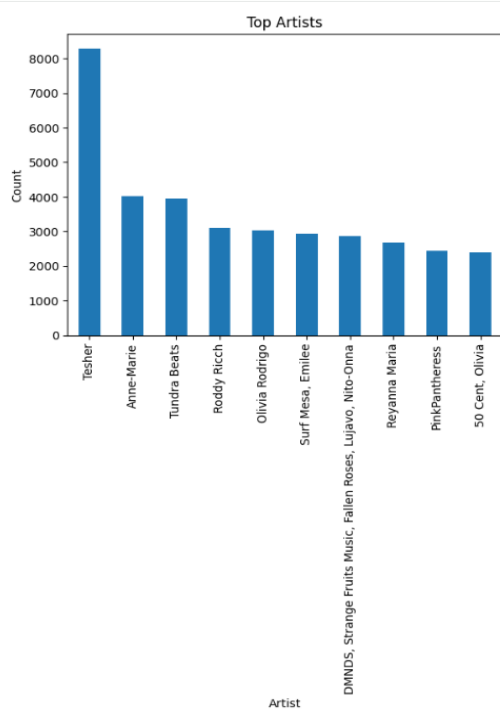


Top Artists

The most frequently accessed artists were identified.

Insights:

- A small number of artists generated significant engagement.
- Popular artists can be prioritized in recommendation strategies.

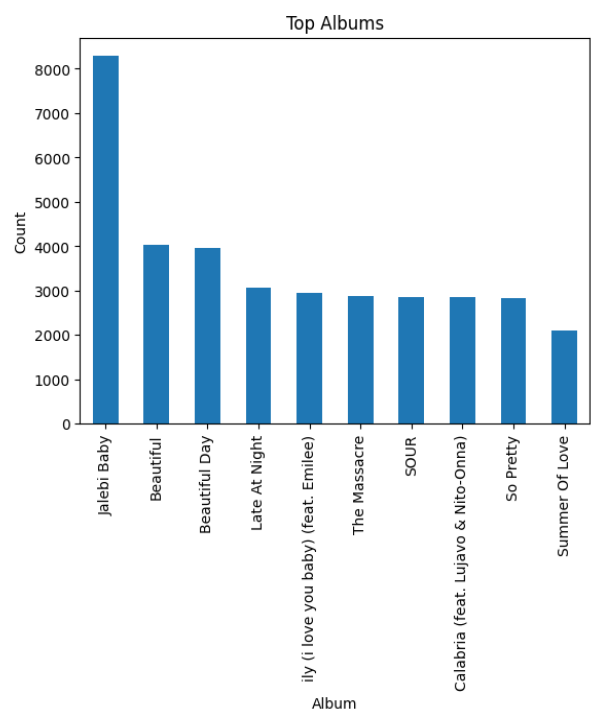


Top Albums

Album popularity was analyzed using event counts.

Insights:

- Certain albums consistently attract higher user engagement.
- Popular albums can be highlighted for increased visibility.



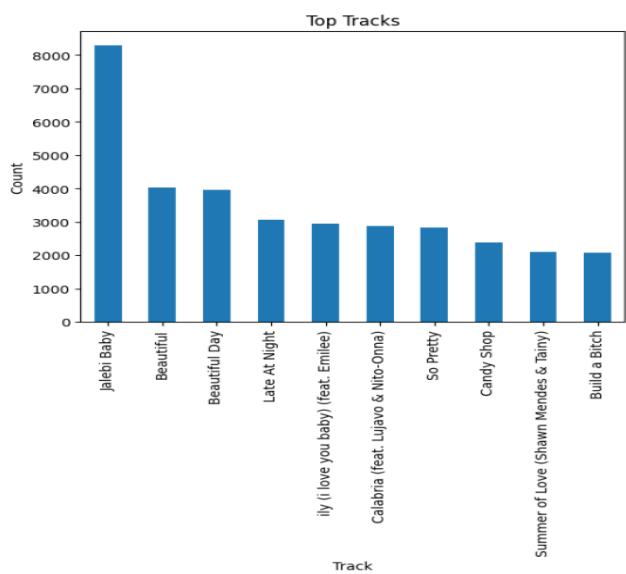
Top Tracks

Track-level analysis was performed to identify the most popular tracks.

Insights:

- Top-performing tracks receive significantly more interactions.
- These tracks can be used for promotional campaigns.

(Insert Top Tracks Graph Here)

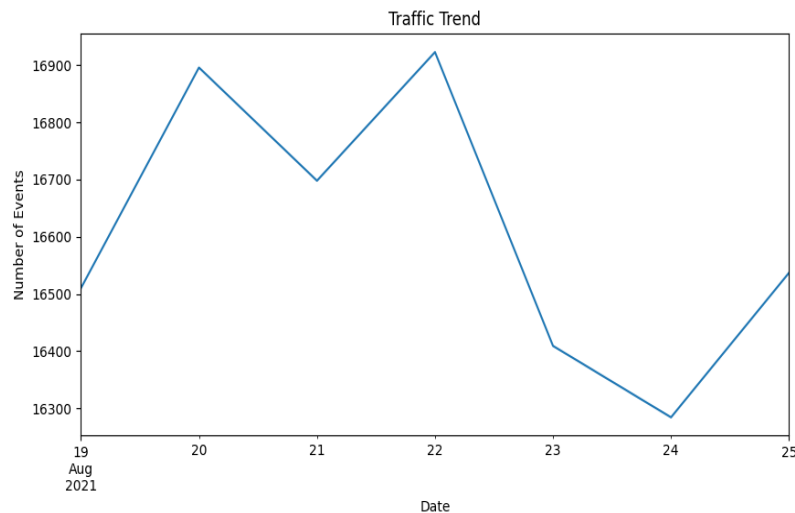


Traffic Trend Over Time

A line chart was created to visualize traffic patterns over time.

Insights:

- User activity varies across different dates.
- Peak traffic periods indicate higher engagement levels.
- Traffic trends can support campaign planning and resource allocation.



Tools Used

- Pandas
- NumPy
- Matplotlib
- Google Colab

Key Findings

The analysis revealed several important insights:

- Traffic is concentrated in a limited number of countries.
- Certain artists, albums, and tracks generate significantly higher engagement.
- User interactions are dominated by specific event types.
- Traffic levels fluctuate over time.
- Popular content drives a large share of total activity.

Recommendations for Alfido Tech:-

1. Focus on High-Traffic Regions

Increase marketing efforts and localization strategies in countries generating the highest traffic.

2. Promote Popular Content

Highlight top-performing artists, albums, and tracks through recommendations and featured sections.

3. Optimize Engagement During Peak Traffic Periods

Schedule campaigns and content releases during high-traffic periods to maximize user engagement.

Conclusion:-

This project successfully analyzed website traffic data using Python and visualization techniques. The analysis provided valuable insights into user activity, geographical distribution, content popularity, and traffic trends.

The findings indicate that focusing on high-performing regions, promoting popular content, and leveraging peak traffic periods can significantly improve user engagement and business performance. These recommendations can help Alfido Tech make informed decisions and enhance overall platform effectiveness.

References:-

- Python Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Website Traffic Dataset

Screenshots and Visualizations:-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('/content/traffic.csv')
df.head()
```

```
...
event    date    country    city    artist    album    track    isrc    linkid
0  click  2021-08-21  Saudi Arabia  Jeddah    Teshher  Jalebi Baby  Jalebi Baby  QZNWQ2070741  2d896d31-97b6-4869-967b-1c5fb9cd4bb8
1  click  2021-08-21  Saudi Arabia  Jeddah    Teshher  Jalebi Baby  Jalebi Baby  QZNWQ2070741  2d896d31-97b6-4869-967b-1c5fb9cd4bb8
2  click  2021-08-21    India  Ludhiana    Reyanna Maria  So Pretty  So Pretty  USUM72100871  23199824-9cf5-4b98-942a-34965c3b0cc2
3  click  2021-08-21    France  Unknown  Simone & Simaria, Sebastian Yatra  No Llores Más  No Llores Más  BRUM72003904  35573248-4e49-47c7-af80-08a960fa74cd
4  click  2021-08-21  Maldives    Malé    Teshher  Jalebi Baby  Jalebi Baby  QZNWQ2070741  2d896d31-97b6-4869-967b-1c5fb9cd4bb8
```

```
print("Rows and Collumn:",df.shape)

print("\nColumns:")
print(df.columns)

print("\nMissing Values:")
print(df.isnull().sum())

... Rows and Collumn: (226278, 9)

Columns:
Index(['event', 'date', 'country', 'city', 'artist', 'album', 'track', 'isrc',
      'linkid'],
      dtype='object')
```

```
Missing Values:
event      0
date       0
```

```
Missing Values:
event      0
date       0
country    11
city       11
artist     37
album       5
track       5
isrc       7121
linkid      0
dtype: int64
```

```
df.drop_duplicates(inplace=True)

print("Shape after removing duplicates:")
print(df.shape)
```

```
Shape after removing duplicates:
(116257, 9)
```

```
print("Total Events:",len(df))

Total Events: 116257
```

```
print("Unique Countries:",df['country'].nunique())
```

```
... Unique Countries: 211
```

```
print("Unique Cities:",df['city'].nunique())
```

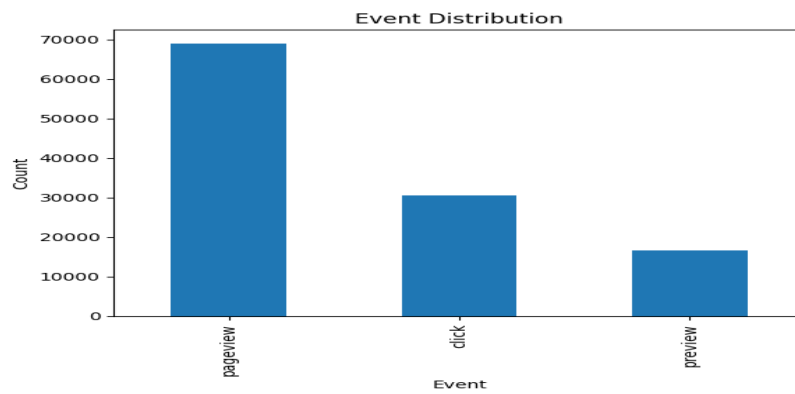
```
Unique Cities: 11690
```

```
df['event'].value_counts().plot(kind='bar')

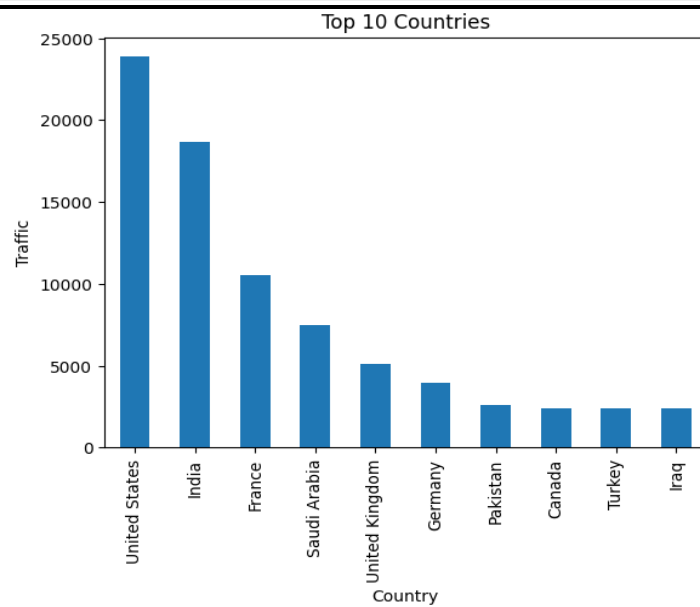
plt.title("Event Distribution")
plt.xlabel("Event")
plt.ylabel("Count")
```



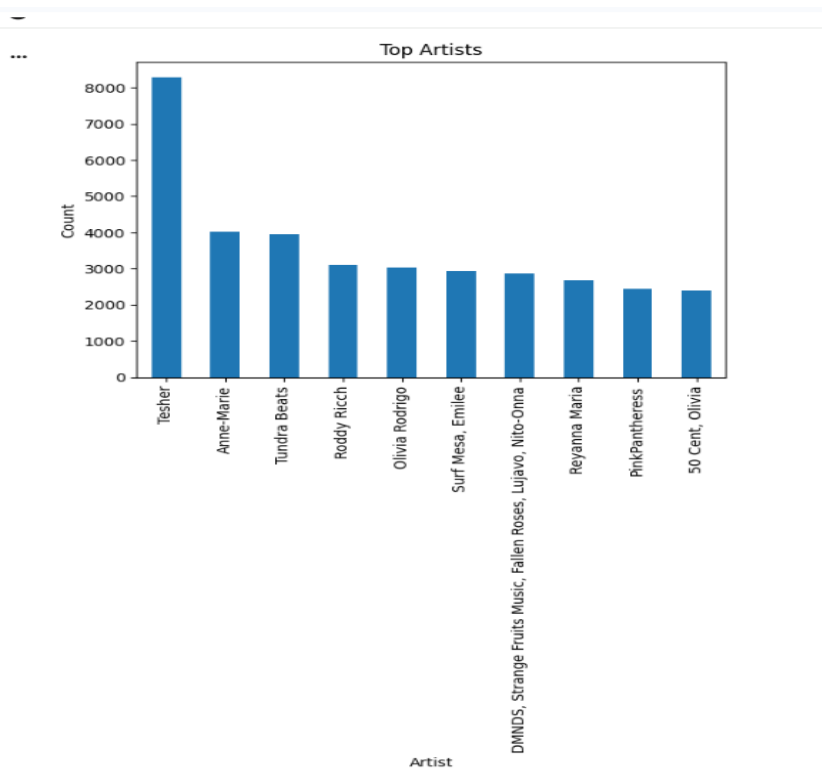
```
plt.xlabel("Event")
plt.ylabel("Count")
plt.show()
```



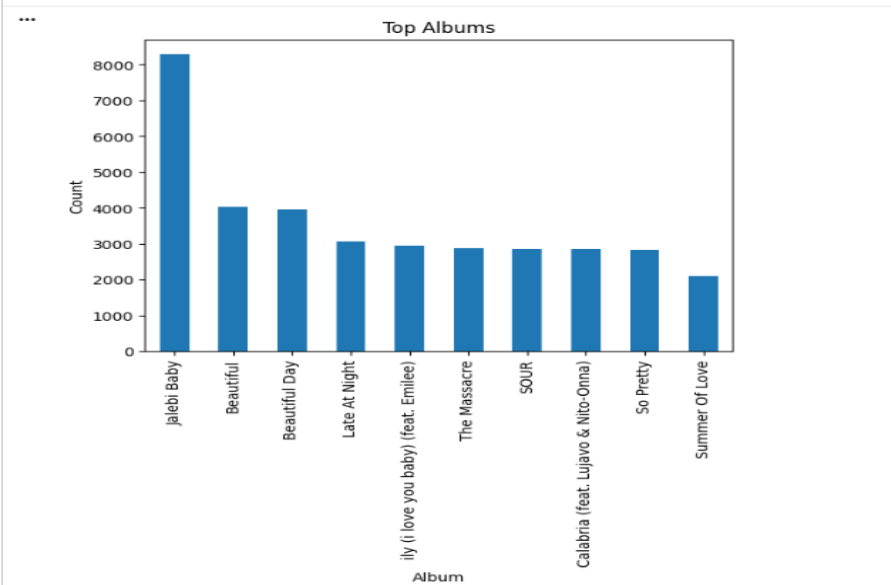
```
df['country'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 Countries")
plt.xlabel("Country")
plt.ylabel("Traffic")
plt.show()
```

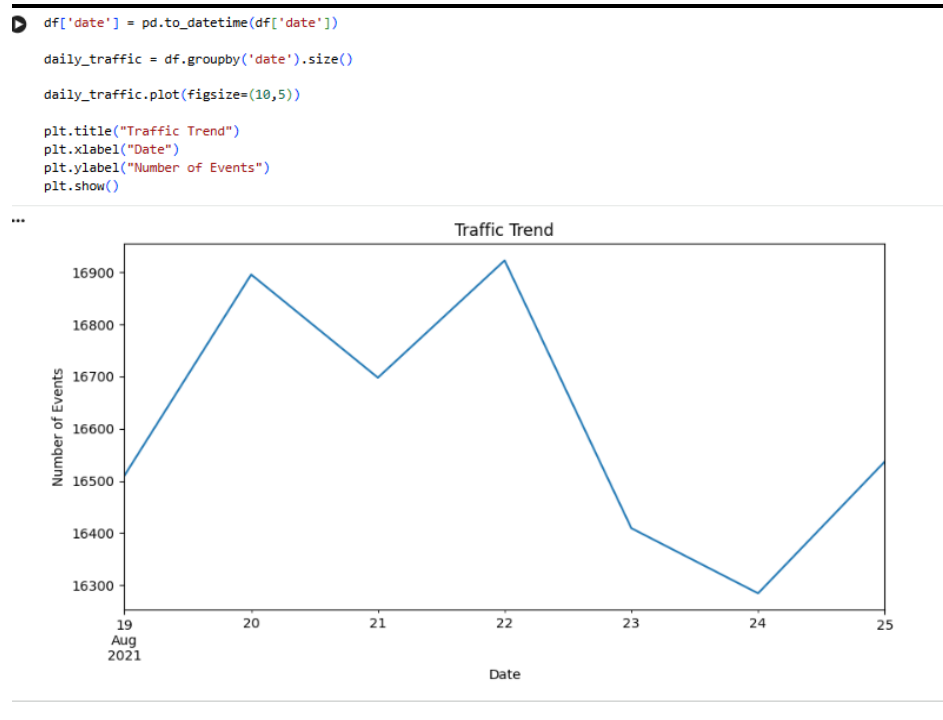
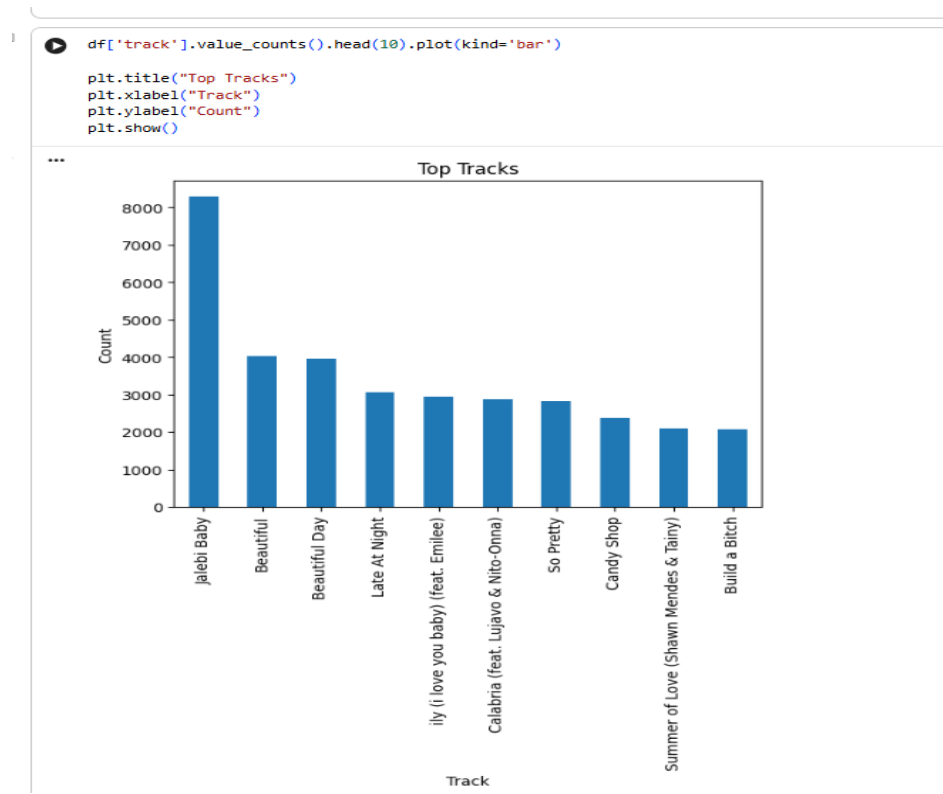


```
df['artist'].value_counts().head(10).plot(kind='bar')
plt.title("Top Artists")
plt.xlabel("Artist")
plt.ylabel("Count")
plt.show()
```



```
df['album'].value_counts().head(10).plot(kind='bar')  
  
plt.title("Top Albums")  
plt.xlabel("Album")  
plt.ylabel("Count")  
plt.show()
```





Code Link:- <https://github.com/Shadnmnn34/InternSpark--InternShip/tree/09c192396d3ad78a25f906ce68e4e83fdd914963/Website%20traffic%20analysis>

